CS440 - Homework 1: Search, MDP, RL

Out: February 6, 2026
Due: February 17, 2026 @ 23:59 pm, submit via Gradescope

Instructions

- This is an **individual** homework. Please write up your solution independently. You can discuss with your classmates, and if so, please list their names on the front page of your write-up.
- Please list the number of late days you used (max 2).
- You may consult the course slides and assigned reading without citing it. Please refrain from using other sources. If absolutely necessary, cite the sources you consulted.
- Please write convincingly but not excessively.
- Please type your solutions if possible. We will give 3% bonus (of the total homework score) to typed solution. You can find a LaTeX template on Canvas under the "homework" folder.
- Please start early, and have fun!

1 Search: Museum Patrol (11pts)

A Museum guard is tasked with checking on several exhibits in a museum gallery. In order to use its time efficiently, the guard should only visit each location once and should not backtrack to exhibits it has already visited. The following matrix shows the distance between each exhibit, e.g., distance between A and C would be 3, "–" indicates no path, and paths are *not* bidirectional. *Note: When performing a search, you can assume that ties are settled by the bot in alphabetical order.*

	S	A	B	C	D	E	F	G
S	–	2	5	–	–	–	–	–
A	–	–	–	3	–	–	6	–
B	–	–	–	2	–	4	–	–
C	–	–	–	–	9	7	–	–
D	–	–	–	–	–	–	–	2
E	–	–	–	–	–	–	–	3
F	–	–	–	–	–	–	–	1

(a) (3pts) Draw a state-space diagram (i.e., search graph) that represents the guard traversing these exhibits.

(b) (5pts) Find a path from S to G for the guard to follow, using breadth first search (BFS), depth first search (DFS), and uniform cost search (UCS).

- Show the path found using each algorithm.
- Calculate the cost (i.e., distance) of the found path for each algorithm. Which algorithm(s) find the optimal path, i.e., the path of lowest cost?

- Which algorithm(s) find a path the most quickly, i.e., the path that expands the fewest number of states?

(c) (3pts) Consider a heuristic function that maps each state to a heuristic value defined as follows:

- $h(S) = 7, h(A) = 5, h(B) = 6, h(C) = 4, h(D) = 3, h(E) = 2, h(F) = 5$

Find a path and its associated cost (i.e., distance) for the guard to follow, using both Greedy search and A* search. How do the paths found by these algorithms compare to the optimal path found in part (b)? Why might they differ? (Hint: Is the heuristic admissible? Is the heuristic consistent?)

2 MDP: Self-driving Car (13pts)

You are developing a self-driving vehicle that transports passengers from home (position 1) to school (position n). At each time step, the car has a current position $x \in \{1, \ldots, n\}$ and a current velocity $v \in \{0, \ldots, m\}$. The car always starts with $v = 0$, and at each time step, the car can either increase the velocity by 1, decrease it by 1, or keep it the same; this new velocity is used to advance x to a new position. The velocity is not allowed to exceed the speed limit m nor return to 0.

To ensure that individuals do not drive recklessly, the university has installed speed bumps at a select subset of the n locations, i.e., the speed bumps are located at $B \subseteq \{1, \ldots, n\}$. The car is not permitted to enter, leave, or pass over a speed bump with velocity more than $k \in \{1, \ldots, m\}$. *Your goal is to arrive at position n with velocity 1 in the shortest time (smallest number of time steps).*

Example: Figure 1 shows $n = 9$ positions and one speed bump $B = \{5\}$. If the maximum speed is $m = 3$ and $k = 1$ for a speed bump, then a legal path can be the following:

$$(1,0) \xrightarrow{+1} (2,1) \xrightarrow{+1} (4,2) \xrightarrow{-1} (5,1) \xrightarrow{0} (6,1) \xrightarrow{+1} (8,2) \xrightarrow{-1} (9,1)$$

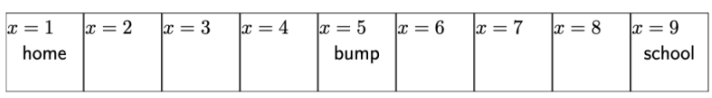

Figure 1: A valid path that takes 6 time steps with $m = 3$ and $k = 1$. The position-velocity pairs (x, v) at each time step are illustrated. Each number above the arrow indicating that action, i.e., the acceleration (change in velocity).

(a) (3pts) *For simplicity, let us assume that there are no speed bumps along the route.* It turns out that something is wrong with the car's braking system. As a result, whenever you attempt to decrease the velocity by 1, there is a failure probability of α, meaning that the velocity remains unchanged instead. Assume a reward of R for reaching the school at a velocity of 1 and a cost of 1 per time step.

Let's now formulate this problem as an MDP:

- $s_{\text{start}} = (1, 0)$
- $\text{Actions}((x, v)) \in \{+1, 0, -1\}$
- $T((x', v')|(x, v), a) =?$ *(to be filled out)*

- $\text{Reward}((x', v')|(x, v), a) = R \cdot \mathbf{1}[x' = n \wedge v' = 1] - 1$, where $\mathbf{1}[\cdot]$ is an indicator function, meaning a reward of R is achieved only when the car reaches school at the velocity of 1.
- $\text{IsEnd}((x, v)) = \mathbf{1}[x \geq n]$

Fill out the transition probabilities T by taking into account the possible brake failure. *(Hint: express x' and v' in terms of x, v, and a.)*

(b) Let us examine the impact of unreliable brakes. Consider the scenario depicted in Figure 2.

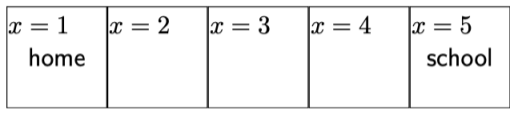

Figure 2: A small driving environment without speed bumps.

Consider two policies:

- π_1: always move with velocity 1, that is

$$\pi_1((1,0)) = +1 \quad \pi_1((2,1)) = 0 \quad \pi_1((3,1)) = 0 \quad \pi_1((4,1)) = 0$$

- π_2: speed up and slow down:

$$\pi_2((1,0)) = +1 \quad \pi_2((2,1)) = +1 \quad \pi_2((4,2)) = -1$$

(i) (2pts) Compute the expected utility of π_1 as a function of α and R (with discount $\gamma = 1$).

(ii) (2pts) Compute the expected utility of π_2 as a function of α and R (with discount $\gamma = 1$).

(iii) (2pts) For what values of α and R does π_2 obtain higher expected reward than π_1? Your answer should be an expression relating α and R.

(c) (4pts) Let's return to the original driving environment, now extended to $n = 11$ positions with a speed bump located at $B = \{5\}$ as shown in Figure 3.

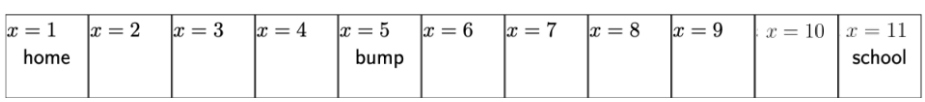

Figure 3: An extended example of the driving setting from Figure 1.

We will start with the cautious policy π_1 from part (b):

$$\pi_1((1,0)) = +1, \qquad \pi_1((x,v)) = 0 \quad \text{for all other states } s = (x,v)$$

i.e., the car maintains velocity 1 until it reaches the school.

Starting from π_1, **use policy iteration** to determine π_2 and evaluate π_2 (i.e., calculating $V_{\pi_2}(s)$). Assume a terminal reward $R = 15$ for reaching position 11 at velocity $v = 1$. Assume a brake failure of probability $\alpha = 0.1$, a maximum velocity of $m = 3$, a speed limit of $k = 1$, a cost of 1 per time step, and a discount factor $\gamma = 1$. You receive a reward of 0 if you enter the speed bump or school at a velocity greater than the specified limit $k = 1$.

Hint: (1) You should first perform a policy evaluation on π_1, and then work on policy improvement. (2) For policy improvement, check the states where accelerating could plausibly save time while still allowing the car to brake before the next constraint.

3 RL: Riding the Bus (15pts)

Christopher needs to travel from their home, situated at position 1, to the gym, located at position n. At each point s, they have two choices: (i) they can walk to the next spot $s + 1$, which takes 1 unit of time, or (ii) wait for a bus that may arrive with a probability of ϵ. If the bus arrives, it will take them to the gym in $1 + \alpha(n - s)$ units of time, where α is a given parameter. If the bus does not show up, they remain in the same place, which also takes 1 unit of time. *Note: the objective is to minimize the time taken to reach the gym, which can be viewed as maximizing negative time, or "reward."*

1	2	3	4	...	n
House				...	Gym

Figure 4: Christopher riding the bus.

(a) (3pts) We have formalized the problem as an MDP for you:

- State: $s \in \{1, 2, \ldots, n\}$ is Christopher's location
- $\text{Actions}(s) = \{\text{Walk, Bus}\}$
- $\text{Reward}(s, \text{Walk}, s') = \begin{cases} -1 & \text{if } s' = s + 1 \\ -\infty & \text{otherwise} \end{cases}$
- $\text{Reward}(s, \text{Bus}, s') = \begin{cases} -1 - \alpha(n - s) & \text{if } s' = n \\ -1 & \text{if } s' = s \\ -\infty & \text{otherwise} \end{cases}$
- $T(s'|s, \text{Walk}) = \begin{cases} 1 & \text{if } s' = s + 1 \\ 0 & \text{otherwise} \end{cases}$
- $T(s'|s, \text{Bus}) = \begin{cases} \epsilon & \text{if } s' = n \\ 1 - \epsilon & \text{if } s' = s \\ 0 & \text{otherwise} \end{cases}$
- $\text{IsEnd}(s) = \mathbf{1}[s = n]$

Now, calculate $V_{walk}(s)$ and $V_{bus}(s)$, i.e., the value for the "always walk" policy and the "always wait for the bus" policy. Use the provided variables ϵ, α, n, and s in your expression. Assume a discount factor of $\gamma = 1$.

(b) (5pts) Due to administrative inefficiencies, Christopher's town is unable to supply them with the bus schedule (i.e., the transition probabilities or a reward function). To work around this issue, Christopher decides to apply reinforcement learning—specifically, the Q-learning algorithm—to figure out the optimal strategy. They begin navigating the town, both by walking and taking the bus, while collecting the following data:

s_0	a_1	r_1	s_1	a_2	r_2	s_2	a_3	r_3	s_3	a_4	r_4	s_4	a_5	r_5	s_5
1	Bus	-1	1	Bus	-1	1	Bus	3	3	Walk	1	4	Walk	1	5

Run the Q-learning algorithm *once over the given data* to compute an estimate of the optimal Q-value $Q_{\text{opt}}(s, a)$. Process the episodes from left to right. Assume all Q-values are initialized to zero, and use a learning rate of $\eta = 0.5$ and a discount of $\gamma = 1$. Specifically, please fill in the following Q-learning updates on each (s, a, r, s'):

- On (s_0, a_1, r_1, s_1), $\hat{Q}(1, Bus) =?$
- On (s_1, a_2, r_2, s_2), $\hat{Q}(1, Bus) =?$
- On (s_2, a_3, r_3, s_3), $\hat{Q}(1, Bus) =?$
- On (s_3, a_4, r_4, s_4), $\hat{Q}(3, Walk) =?$
- On (s_4, a_5, r_5, s_5), $\hat{Q}(4, Walk) =?$

(c) Christopher's employer has decided to start subsidizing their bus pass. As a result, Christopher chooses to take the bus when they are at $s < \frac{3}{4}n$, and for states $s \geq \frac{3}{4}n$ they will walk.

(i) (1pt) Write down a policy function $\pi(s)$ to represent this.

(ii) (2pts) Christopher is curious about the expected duration of their commute under this policy. Would you suggest Christopher to use Q-Learning? Why or why not?

(d) As illustrated in Figure 4, suppose that Christopher starts from their house (state 1) and wants to arrive at the gym (assuming state $n > 4$). Christopher would like to use TD-learning by following the policy in part (c) to find the values for each state, i.e.,

$$V^{\pi}(s) \leftarrow V^{\pi}(s) + \alpha[R(s, \pi(s), s') + \gamma V^{\pi}(s') - V^{\pi}(s)]$$

(i) (2pts) Will the algorithm eventually converge to the correct value function $\hat{V}_{\pi}(s)$ for *all states* after Christopher makes enough trips to the gym? Why might there be a potential issue with convergence in this case?

(ii) (1pt) How can Christopher modify their strategy to a new policy π' to ensure that the estimated value function $\hat{V}_{\pi'}(s)$ converges across all states?

(e) (1pt) Reflecting on their previous attempt at Q-learning from part (b), Christopher wants to collect more data *while still reaching the gym in a reasonable amount of time for each trip.* What type of policy would allow them to do this while ensuring convergence?